L2.6a Optimization

§3.7 — sort the question, then run the template

Optimization under a constraint

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1. Find two numbers whose difference is 100 and whose product is a minimum.

2. The rate P at which a type of plankton makes food by photosynthesis depends on the light intensity I (in light units) by

$$P = \frac{100I}{I^2 + I + 4}.$$

For what light intensity is P a maximum?

The optimization template

Solve each problem with all five steps of the optimization template. Answer the question that was asked, with units.

3. A farmer wants to fence in a rectangular plot of land adjacent to the north wall of his barn. No fencing is needed along the barn, and the fencing along the west side of the plot is shared with a neighbor who will split the cost of that portion of the fence. The fencing costs \$20 per linear foot to install, and the farmer is not willing to spend more than \$5000. Find the dimensions of the plot that would enclose the most area.

4. Let

$$f(x) = \frac{kx}{1+x^2}, \qquad x > 0,$$

where k is a positive constant. Find the value of x where f has its maximum, and the maximum value.

Stop. Look at the interval before you commit to an answer. If it has no ends, what settles whether a critical number is a Max or a min — and what is the name of that test? Here x is the variable and k only rides along.

5. The top and bottom margins of a poster are each 6 cm and the side margins are each 4 cm. The area of printed material on the poster is fixed at 384 cm^2. Find the dimensions of the poster with the smallest area.

Stop. Which rectangle is the 384 cm^2 one — the poster, or the printed part? Label both on your picture before you write the optimization equation (OEq).